

Chemistry Interim · Practice Exam 1

Chemistry · Unit 1 Atomic Structure (60%) · Unit 2 Periodic Trends (20%) · Unit 3 Chemical Bonding (20%)

Name: _____

Date: _____

Period: _____

Time: 60 minutes. **Total:** 80 points. Part A: 18 multiple-choice questions, 2 points each (36). Part B: 5 free-response questions (44). Weighting: Unit 1 Atomic Structure 60%, Unit 2 Periodic Trends 20%, Unit 3 Chemical Bonding 20%. Show your work in Part B.

Reference. Average atomic mass = sum of (isotope mass × fractional abundance). Electronegativity: H 2.1 · C 2.5 · N 3.0 · O 3.5 · F 4.0 · Na 0.9 · Mg 1.2 · Al 1.5 · Si 1.8 · P 2.1 · S 2.5 · Cl 3.0 · K 0.8 · Ca 1.0 · Br 2.8. Bond type from ΔEN : below 0.5 nonpolar covalent · 0.5 to 1.7 polar covalent · above 1.7 ionic. A periodic table and a calculator may be used.

Part A · Multiple Choice (2 points each; circle one answer)

- Which scientist discovered the electron, using a cathode ray tube?
A Dalton
B Thomson
C Rutherford
D Chadwick
- In the gold foil experiment, most alpha particles passed straight through the foil. This showed that
A atoms are solid spheres
B the nucleus is negative
C electrons move in fixed orbits
D most of the atom is empty space
- A limitation of Bohr's model is that it
A has no nucleus
B cannot explain why atoms are neutral
C works only for atoms with one electron, such as hydrogen
D says that atoms cannot be divided
- Which particle has a relative mass of about 1 amu and no charge?
A neutron
B proton
C electron
D alpha particle
- How many neutrons are in an atom of $^{39}_{19}\text{K}$?
A 19
B 39
C 58
D 20
- An ion has 16 protons, 16 neutrons and 18 electrons. Its symbol is
A $^{32}_{16}\text{S}^{2+}$
B $^{32}_{16}\text{S}^{2-}$
C $^{34}_{16}\text{S}^{2-}$
D $^{32}_{18}\text{Ar}$

7. Two atoms are isotopes of the same element if they have
- A the same number of neutrons but different numbers of protons
 - B the same mass number
 - C the same number of protons but different numbers of neutrons
 - D different numbers of electrons
8. An element has two isotopes: X-10 (20%) and X-11 (80%). Its average atomic mass is closest to
- A 10.2 amu
 - B 10.5 amu
 - C 10.8 amu
 - D 11.0 amu
9. In nuclear fusion,
- A a heavy nucleus splits into smaller nuclei
 - B two light nuclei combine to form a heavier nucleus
 - C electrons are transferred from one atom to another
 - D an electron falls to a lower energy level
10. What is the electron configuration of a neutral phosphorus atom ($Z = 15$) in its ground state?
- A $1s^2 2s^2 2p^6 3s^2 3p^3$
 - B $1s^2 2s^2 2p^6 3s^2 3d^3$
 - C $1s^2 2s^2 2p^8 3s^2 3p^1$
 - D $1s^2 2s^2 2p^6 3s^1 3p^4$
11. An atom emits light when an electron
- A absorbs energy
 - B jumps to a higher energy level
 - C leaves the atom
 - D falls from a higher energy level to a lower one
12. Which atom has the largest atomic radius?
- A Na
 - B Mg
 - C Al
 - D Cl
13. Moving down group 1, the first ionization energy
- A increases, because there are more protons
 - B decreases, because the outer electron is farther from the nucleus and more shielded
 - C stays the same
 - D decreases, because the atoms have fewer protons
14. Which element is the most electronegative?
- A fluorine
 - B oxygen
 - C chlorine
 - D cesium
15. Compared with a neutral chlorine atom, a chloride ion (Cl^-) is
- A smaller, because it has gained a proton
 - B the same size
 - C smaller, because it has more electrons
 - D larger, because the extra electron adds repulsion while the nuclear charge stays the same

16. Which compound contains ionic bonds?

- A** CO₂
C MgO

- B** H₂O
D CH₄

17. What is the name of FeCl₃?

- A** iron chloride
C iron trichloride

- B** iron(III) chloride
D iron(I) chloride

18. Which molecule is nonpolar?

- A** CO₂
C NH₃

- B** H₂O
D HCl

Part B · Free Response (show your work)

19. Atomic models (8 points) Unit 1

(a) Complete the table. (4 points)

Scientist	Experiment or evidence	What was discovered or proposed
Thomson		
Rutherford		
Bohr		
Chadwick		

(b) State two observations from the gold foil experiment and the conclusion drawn from each. (2 points)

(c) State one problem with Rutherford's model, and explain how Bohr's model solved it. (2 points)

20. Atoms, isotopes and average atomic mass (10 points) Unit 1**(a)** Complete the table. (4 points)

Species (symbol with mass number, atomic number and charge)	Protons	Neutrons	Electrons
${}^{27}_{13}\text{Al}^{3+}$			
	17	18	18
${}^{15}_7\text{N}$			
a neutral atom of potassium-41			

(b) Magnesium has three isotopes. Calculate its average atomic mass. (3 points)

Isotope	Mass (amu)	Abundance (%)
Mg-24	23.985	78.99
Mg-25	24.986	10.00
Mg-26	25.983	11.01

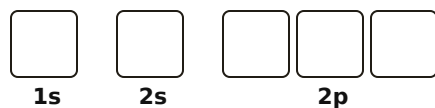
Work:

(c) Chlorine has two isotopes: Cl-35 (34.969 amu) and Cl-37 (36.966 amu). Its average atomic mass is 35.453 amu. Find the percent abundance of each isotope. (3 points)

Work:

21. Electrons and light (8 points) Unit 1**(a)** Write the full electron configuration and the noble gas configuration of calcium ($Z = 20$). (2 points)

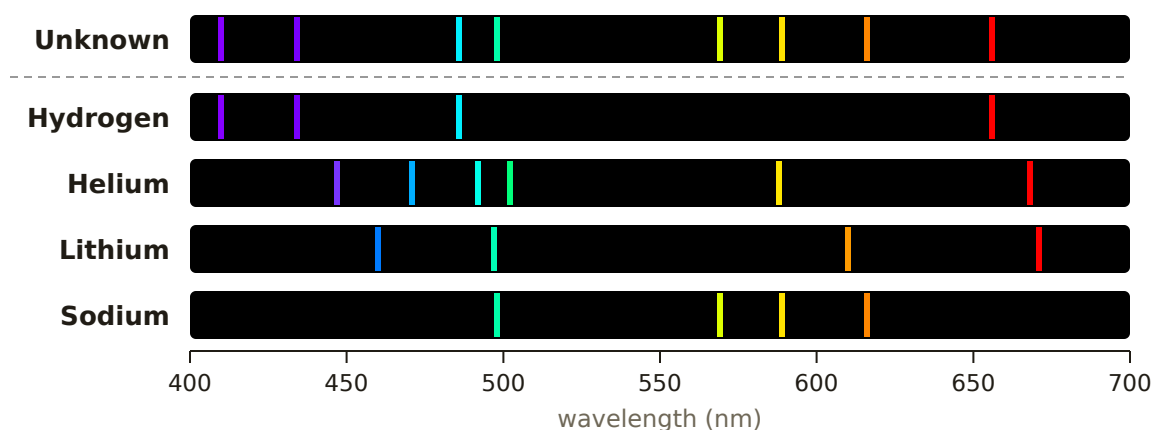
(b) Complete the orbital filling diagram for oxygen ($Z = 8$), and state how many unpaired electrons it has. (2 points)



Unpaired electrons: _____

(c) A neutral atom has the configuration $1s^2 2s^2 2p^6 3s^1 3p^1$. Identify the element and state whether the atom is in the ground state or an excited state. Explain. (2 points)

(d) The emission spectrum of an unknown gas mixture is shown with four reference spectra. Which of the four elements are in the unknown? Explain how you decided. (2 points)



22. Periodic trends (8 points) Unit 2

(a) Arrange Mg, S, Na and Cl in order of increasing atomic radius. (2 points)

(b) Explain why atomic radius decreases from left to right across period 3. (2 points)

(c) Which has the higher first ionization energy, Li or K? Explain. (2 points)

(d) Which is larger, a sodium atom (Na) or a sodium ion (Na⁺)? Explain. (2 points)

23. Chemical bonding (10 points) Unit 3

(a) Complete the table. (3 points)

Bond	Δ EN (show the subtraction)	Type of bond
Na-F		
O-H (in water)		
Cl-Cl		

(b) Name each compound. (3 points)

Formula	Name
CaBr ₂	
CuO	
N ₂ O ₅	

(c) Draw the Lewis structure of PCl₃, showing every lone pair. Then state its electron geometry, its molecular geometry, and whether the molecule is polar or nonpolar. (4 points)

Lewis structure:

Electron geometry, molecular geometry, polarity: